



E-Commerce Sales & Business Analytics

1. Executive Summary

ElectroHub is an end-to-end e-commerce analytics project designed to turn transactional and operational data into management-ready business decisions. The project examines four connected areas: overall financial performance, customer and growth behavior, product profitability, and delivery/operational performance.

The analysis starts with five related tables—customers, orders, order_items, products, and delivery. The data was inspected and cleaned before analysis. Python was then used for exploratory data analysis, SQL for structured and repeatable business questions, and Power BI for an interactive four-page management dashboard.

Core Business Problem

How can ElectroHub increase sales and profit while retaining valuable customers and reducing the business impact of poor delivery, cancellations, returns, and inefficient discounting?

Headline Business KPIs

KPI	Result Shown in Dashboard	Why Management Cares
Total Sales	≈ ₹1.6 Billion	Measures the overall revenue scale.
Total Profit	≈ ₹460.32 Million	Measures the earnings generated after the modeled costs.
Total Orders	≈ 65K	Shows transaction volume and demand activity.
Total Customers	≈ 15K	Shows the size of the customer base.
Average Order Value	≈ ₹25.02K	Shows the average revenue generated per order.
Profit Margin	≈ 28.29%	Shows how efficiently sales are converted into profit.

The exact values above are dashboard-level results and should be interpreted together rather than individually. For example, high sales are positive, but management should also check profit margin, customer value, product margin, and operational performance before deciding where to invest.

2. Raw Data Understanding

The raw dataset was divided into five tables. Each table represents a different business entity or process. This structure supports relational analysis and prevents mixing customer-level, order-level, product-level, and delivery-level information.

Table	Important Columns	Business Role
customers	Customer_ID, State, Customer_Tier, Acquisition_Channel, Age, Gender	Describes who the customers are and where they came from.
orders	Order_ID, Customer_ID, Order_Date	Describes customer transactions and order timing.
order_items	Order_ID, Product_ID, Quantity, Unit_Price, Gross_Sales, Discount_Pct,	Provides product-level financial details.

	Discount_Amount, Net_Sales, Cost, Gross_Profit/Profit	
products	Product_ID, Product_Name, Category	Provides product master information and category grouping.
delivery	Order_ID, Warehouse, Promised_Days, Actual_Delivery_Days, Shipping_Cost, Status, Payment_Method	Measures delivery and operational performance.

Raw Data Quality Questions

- Are important IDs unique and usable for joins?
- Are dates stored as dates and numerical measures stored as numeric values?
- Are there missing customer, order, product, or delivery records?
- Are categorical values consistent, such as State, Category, Customer_Tier, Acquisition_Channel, Warehouse, Status, and Payment_Method?
- Are quantities, prices, discounts, sales, profit, and delivery days within reasonable business ranges?
- Could unusual values be genuine transactions rather than data-entry errors?

3. Data Cleaning & Preparation

Cleaning was performed with the objective of making the data analytically reliable without destroying legitimate business information. For each issue, the project considered the affected column/rows, the business consequence, and the appropriate treatment.

Table / Column	Rows / Issue Checked	Problem It Can Cause	Treatment / Decision	Why It Matters
All tables / all rows	Duplicate rows	Duplicates can inflate sales, profit, orders, or customer counts.	Checked duplicate rows and identifiers; genuine duplicates were removed where applicable.	Prevents double counting.
customers / Customer_ID	Missing or duplicate IDs	Customers may be counted incorrectly or fail to join with orders.	Validated IDs and checked uniqueness/referential use.	Customer KPIs depend on reliable IDs.
orders / Order_ID, Customer_ID	Missing/duplicate order identifiers	Order count and customer-order relationships can become incorrect.	Checked identifiers and distinct-order logic.	Ensures accurate order and AOV calculations.
orders / Order_Date	Wrong type or invalid dates	Time trends and monthly analysis can fail.	Converted/validated Order_Date as a date field.	Required for sales trends and time-based SQL/Power BI analysis.
customers / State, Customer_Tier, Acquisition_Channel	Spaces/inconsistent text	The same category can appear as separate groups.	Trimmed and standardized categorical text where needed.	Prevents fragmented customer analysis.
order_items / Quantity	Very low/high quantities	Can distort sales and product rankings if caused by entry errors.	Investigated outliers using business logic before retaining/removing.	Quantity directly affects revenue and profit.
order_items / Unit_Price	Extreme price values	Can create abnormal sales and profit totals.	Checked extreme values against product/business context.	Protects financial KPIs.
order_items / Discount_Pct	Negative values and unusually high discounts (up to ~60% observed)	Invalid discounts can distort Net Sales, Profit, and Margin.	Negative values were treated as suspicious/invalid; high values were investigated rather than automatically deleted.	Discount is directly linked to revenue and profitability.
order_items / Discount_Amount	Very large values (maximum observed was suspiciously high)	Can reduce net sales/profit incorrectly.	Checked relationship with sales, discount %, and quantity before deciding treatment.	Prevents false profitability conclusions.
order_items /	Unexpected or	Sales KPIs can be	Validated calculated	Sales is the foundation for

Gross_Sales, Net_Sales	inconsistent totals	overstated or understated.	relationships between price, quantity, discount, and net sales.	AOV and margin.
order_items / Gross_Profit or Profit	Negative or unusually low profit	May indicate genuine loss-making products or incorrect cost/sales data.	Investigated rather than automatically deleting negative profit.	Negative profit can be a valuable business insight.
delivery / Promised_Days	Observed range around 3–7 days	Extreme values may be incorrectly entered.	Checked whether values were operationally plausible.	Used to determine whether an order was late.
delivery / Actual_Delivery_Days	Values up to about 14 days	Can affect average delivery and late-delivery KPIs.	Retained plausible delays as potentially genuine operational events.	Real delays should remain visible rather than being removed as outliers.
All related tables	Broken/missing relationships	Joins can lose or duplicate business records.	Checked key relationships between Customer_ID, Order_ID, and Product_ID.	Required for reliable SQL joins and Power BI relationships.

Important Cleaning Principle

The project did not follow the rule 'remove every outlier.' An unusual value was first evaluated against business logic. For example, a 14-day actual delivery can be a genuine late delivery, while a negative discount percentage is more likely to violate the intended business rule. This distinction is important because removing genuine business events can hide the exact problems management needs to see.

4. Python Exploratory Data Analysis (EDA)

Python was used after cleaning to explore the dataset and identify the KPIs and patterns that deserved deeper analysis in SQL and Power BI. The focus was on a small set of business-relevant analyses rather than producing many unrelated visualizations.

4.1 Total Sales

```
total_sales = df['Sales'].sum()
```

Why used: Total Sales establishes the overall revenue baseline. It is useful for comparing categories, states, products, and time periods and becomes the base for AOV and profit-margin calculations.

4.2 Total Profit

```
total_profit = df['Profit'].sum()
```

Why used: Sales alone do not show whether the business is financially healthy. Total Profit shows how much earnings are generated from the modeled transactions.

4.3 Profit Margin

```
profit_margin = df['Profit'].sum() / df['Sales'].sum() * 100
```

Why used: Margin measures profitability efficiency. It answers how much profit is generated for every ₹100 of sales, which is more useful for decision-making than looking at profit alone.

4.4 Total Orders and AOV

```
total_orders = df['Order_ID'].nunique()
AOV = total_sales / total_orders
```

Why used: Order volume shows transaction activity, while AOV shows the average revenue contribution of an order. AOV is useful when evaluating upselling, cross-selling, bundles, and customer spending behavior.

4.5 Sales by Category

```
category_sales = df.groupby('Category')['Sales'].sum().sort_values(ascending=False)
```

Why used: Identifies the categories that drive revenue. In the dashboard, Electronics is the largest sales contributor. The next analytical step is to compare category sales with profit and margin, because the highest-selling category is not automatically the most profitable.

4.6 Sales vs Profit — High Sales + Low Profit

```
category_profit = df.groupby('Category')['Profit'].sum().sort_values(ascending=False)
```

Why used: This comparison is central to the project. It identifies products/categories where revenue is strong but profit contribution is weak. Such products should be investigated for discounting, pricing, product cost, or other margin drivers.

4.7 Customer / Acquisition Analysis

```
customer_sales = df.groupby('Customer_Tier')['Sales'].sum().sort_values(ascending=False)
```

```
channel_sales = df.groupby('Acquisition_Channel')['Sales'].sum().sort_values(ascending=False)
```

Why used: Customer counts alone do not show customer value. Combining customer tier, sales, acquisition channel, and AOV helps management understand which segments and channels contribute financially.

4.8 Delivery Performance

```
avg_delivery_days = delivery['Actual_Delivery_Days'].mean()
```

```
late_pct = (delivery['Actual_Delivery_Days'] > delivery['Promised_Days']).mean() * 100
```

Why used: Delivery performance links operations with customer experience. Average delivery time measures speed, while late-delivery percentage measures reliability against the promised service level.

Python EDA → Business Value

Finding Area	Business Question	Decision Supported
Sales	How much revenue are we generating?	Set revenue baseline and compare segments.
Profit / Margin	Are sales translating into earnings?	Focus on profitable growth.
AOV	How much does the average order contribute?	Evaluate upselling and cross-selling.
Category / Product	What drives revenue and profit?	Prioritize strong products and investigate weak margins.
Customer / Channel	Who is valuable and where do they come from?	Improve retention and marketing allocation.
Delivery	Are operational delays occurring?	Target warehouses/states needing improvement.

5. SQL Business Analysis

SQL was used to convert the exploratory findings into structured, reproducible business analysis across the five tables. The key advantage is that SQL can join related tables and answer management questions directly at the correct business grain.

5.1 Total Sales

```
SELECT SUM(Sales) AS Total_Sales
FROM order_items;
```

Why this KPI is useful: Creates the primary revenue baseline against which categories, products, states, customers, and months can be compared.

5.2 Total Profit

```
SELECT SUM(Profit) AS Total_Profit
FROM order_items;
```

Why this KPI is useful: Measures the total modeled profit and prevents the analysis from focusing only on revenue.

5.3 Total Orders

```
SELECT COUNT(DISTINCT Order_ID) AS Total_Orders
FROM orders;
```

Why this KPI is useful: DISTINCTCOUNT logic is important because one order can contain multiple order-item rows. It prevents order volume from being overstated.

5.4 Total Customers

```
SELECT COUNT(DISTINCT Customer_ID) AS Total_Customers
FROM customers;
```

Why this KPI is useful: Measures the size of the unique customer base and supports customer-level KPIs such as repeat-customer rate.

5.5 Average Order Value

```
SELECT
    SUM(oi.Sales) * 1.0 / COUNT(DISTINCT o.Order_ID) AS AOV
FROM orders o
JOIN order_items oi
    ON o.Order_ID = oi.Order_ID;
```

Why this KPI is useful: Shows the average revenue generated per order and provides a useful target for increasing basket value.

5.6 Monthly Sales

```
SELECT
    YEAR(o.Order_Date) AS Year,
    MONTH(o.Order_Date) AS Month,
    SUM(oi.Sales) AS Sales
FROM orders o
JOIN order_items oi
    ON o.Order_ID = oi.Order_ID
GROUP BY YEAR(o.Order_Date), MONTH(o.Order_Date)
ORDER BY Year, Month;
```

Why this KPI is useful: Explains when revenue changes rather than only reporting the overall total. It supports trend monitoring and seasonal planning.

5.7 Category Sales and Profit

```
SELECT
    p.Category,
    SUM(oi.Sales) AS Sales,
    SUM(oi.Profit) AS Profit
FROM order_items oi
JOIN products p
    ON oi.Product_ID = p.Product_ID
GROUP BY p.Category
ORDER BY Sales DESC;
```

Why this KPI is useful: Shows whether revenue-leading categories are also profit-leading categories.

5.8 Top 10 Products by Sales

```
SELECT TOP 10
    p.Product_Name,
    SUM(oi.Sales) AS Sales
FROM order_items oi
JOIN products p
    ON oi.Product_ID = p.Product_ID
GROUP BY p.Product_Name
ORDER BY Sales DESC;
```

Why this KPI is useful: Identifies the products with the largest revenue impact and provides a shortlist for product-level investigation.

5.9 Customer Tier Performance

```
SELECT
    c.Customer_Tier,
    COUNT(DISTINCT c.Customer_ID) AS Customers,
    SUM(oi.Sales) AS Sales
FROM customers c
JOIN orders o
    ON c.Customer_ID = o.Customer_ID
JOIN order_items oi
    ON o.Order_ID = oi.Order_ID
GROUP BY c.Customer_Tier
ORDER BY Sales DESC;
```

Why this KPI is useful: Connects customer volume with financial contribution, allowing management to distinguish large segments from valuable segments.

5.10 Acquisition Channel Performance

```
SELECT
    c.Acquisition_Channel,
    COUNT(DISTINCT c.Customer_ID) AS Customers,
    SUM(oi.Sales) AS Sales
FROM customers c
JOIN orders o
    ON c.Customer_ID = o.Customer_ID
JOIN order_items oi
    ON o.Order_ID = oi.Order_ID
GROUP BY c.Acquisition_Channel
ORDER BY Sales DESC;
```

Why this KPI is useful: Shows which channels contribute customers and sales, supporting more efficient marketing allocation.

5.11 Repeat Customers

```
SELECT COUNT(*) AS Repeat_Customers
FROM (
    SELECT Customer_ID
    FROM orders
    GROUP BY Customer_ID
    HAVING COUNT(DISTINCT Order_ID) > 1
) x;
```

Why this KPI is useful: Measures customer retention behavior by identifying customers who placed more than one order.

5.12 Late Delivery %

```
SELECT
    100.0 * SUM(
        CASE
            WHEN Actual_Delivery_Days > Promised_Days THEN 1
            ELSE 0
        END
    ) / COUNT(*) AS Late_Delivery_Pct
FROM delivery;
```

Why this KPI is useful: Measures operational reliability against the promised delivery time and helps management identify service-level problems.

6. Power BI Dashboard & DAX

Power BI was used as the final decision-support layer. The dashboard does not simply repeat every SQL query. Instead, it groups the analysis into four management-focused pages and uses reusable DAX measures so KPIs respond to filters and slicers.

6.1 DAX Measures

Measure	DAX Formula	Why Used
Total Sales	Total Sales = SUM(order_items[Sales])	Core revenue KPI used across pages.
Total Profit	Total Profit = SUM(order_items[Profit])	Core earnings KPI.
Total Orders	Total Orders = DISTINCTCOUNT(orders[Order_ID])	Counts unique orders and avoids item-level double counting.
Total Customers	Total Customers = DISTINCTCOUNT(customers[Customer_ID])	Counts unique customers.
AOV	AOV = DIVIDE([Total Sales], [Total Orders], 0)	Measures average revenue per order.
Profit Margin %	Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)	Measures profit efficiency.
Repeat Customers	Repeat Customers = COUNTROWS(FILTER(VALUES(orders[Customer_ID]), CALCULATE(DISTINCTCOUNT(orders[Order_ID]) > 1)))	Identifies customers with more than one order.
Repeat Customer %	Repeat Customer % = DIVIDE([Repeat Customers], [Total Customers], 0)	Converts repeat customers into a retention indicator.
Average Discount %	Average Discount % = AVERAGE(order_items[Discount_Pct])	Supports discount/profit analysis.
Average Delivery Days	Average Delivery Days = AVERAGE(delivery[Actual_Delivery_Days])	Measures delivery speed.
Late Deliveries	Late Deliveries = CALCULATE(COUNTROWS(delivery), delivery[Actual_Delivery_Days] > delivery[Promised_Days])	Counts deliveries that missed the promise.
Late Delivery %	Late Delivery % = DIVIDE([Late Deliveries], COUNTROWS(delivery), 0)	Measures delivery reliability.
Cancellation Rate %	Cancellation Rate % = DIVIDE(CALCULATE(COUNTROWS(delivery), delivery[Status] = "Cancelled"), COUNTROWS(delivery), 0)	Measures order loss from cancellations.
Return Rate %	Return Rate % = DIVIDE(CALCULATE(COUNTROWS(delivery), delivery[Status] = "Returned"), COUNTROWS(delivery), 0)	Measures post-purchase return activity.
Total Shipping Cost	Total Shipping Cost = SUM(delivery[Shipping_Cost])	Shows logistics cost burden.

6.2 Dashboard Page Design

Page	Business Question	Main KPIs / Visuals	Management Value
1. Executive Overview	How is the overall business performing?	Sales, Profit, Orders, Customers, AOV, Margin, Sales	Provides a one-screen management view of business

		Trend, Category Sales, State Sales	health.
2. Customer & Growth Analysis	Who are our valuable customers and which channels drive growth?	Customers by State/Tier, Sales by Tier, Sales & AOV by Channel, Repeat Customers	Supports retention, segmentation, and marketing decisions.
3. Product & Profitability Analysis	Which products/categories drive revenue and profit?	Sales, Profit, Margin, Top 10 Products, Discount vs Profit, High Sales + Low Profit	Prevents revenue-only decisions and highlights margin opportunities.
4. Operations & Delivery	Are delivery and operational problems affecting the business?	Delivery Days, Late %, Warehouse, State, Cancellation/Return, Shipping Cost, Payment Method	Helps management target operational weaknesses.

6.3 Why Power BI Adds Value After SQL

- SQL produces reliable, structured business metrics; Power BI makes those metrics interactive and easy to monitor.
- DAX measures can respond dynamically to slicers such as year, state, category, and other business dimensions.
- Management can move from a high-level KPI to a segment such as a category, customer tier, state, warehouse, or acquisition channel.
- The four-page structure separates financial performance, customer growth, profitability, and operations instead of putting every metric on one crowded page.

7. Business Problems Solved

Business Problem	Analysis Used	How the Dashboard Helps
Overall performance is unclear	Sales, Profit, Orders, Customers, AOV, Margin, Trend	Provides a management-level performance snapshot.
Revenue may hide weak profitability	Profit, Margin, Category/Product Sales vs Profit	Identifies high-revenue but low-profit areas.
Customer value is unclear	Tier, Repeat Customers, AOV, State	Separates customer volume from customer value.
Marketing channel efficiency is unclear	Customers, Sales, AOV by Acquisition Channel	Shows which channels contribute valuable business.
Discounting may reduce earnings	Discount %, Discount vs Profit	Helps investigate margin erosion from promotions.
Delivery issues may affect customer experience	Average Days, Late %, Warehouse, State	Highlights operational areas needing investigation.
Returns and cancellations reduce realized business	Cancellation Rate, Return Rate	Creates operational/customer-experience monitoring KPIs.

8. Business Insights to Present

The final presentation should emphasize findings that lead to decisions. The following are the key insight themes supported by the dashboard; exact percentages or product names should be stated from the final filtered dashboard values.

- Electronics is the largest sales-contributing category in the Executive Overview, making it a major revenue driver.
- The dashboard's sales and profit views should be compared together to identify whether the largest revenue categories are also the strongest profit contributors.

- Customer tier analysis should be used to identify segments that contribute disproportionately high sales relative to their customer count.
- Acquisition-channel sales and AOV together reveal whether a channel is merely generating customers or generating higher-value orders.
- Repeat-customer analysis provides a direct indicator of retention and should be monitored alongside customer acquisition.
- High-sales/low-profit products deserve investigation before additional promotion because they may be generating revenue without attractive margins.
- Late-delivery and warehouse analysis identifies operational bottlenecks that can affect customer experience, cancellations, and returns.

9. Business Recommendations

1. Prioritize profitable growth: evaluate sales and profit together before increasing inventory or marketing spend.
2. Investigate high-sales/low-profit products: review product cost, selling price, discount levels, and contribution margin.
3. Optimize discounting: use targeted discounts where they increase order value or conversion without unnecessarily reducing margin.
4. Invest in high-value acquisition channels: compare customer volume, sales, and AOV before reallocating marketing budget.
5. Strengthen customer retention: use loyalty benefits, personalized offers, and post-purchase engagement to increase repeat purchases.
6. Improve weak warehouse/state performance: investigate processing time, inventory availability, and logistics partners where delivery time is consistently high.
7. Monitor cancellations and returns together with delivery performance: a rise in these KPIs can signal customer-experience or operational problems.
8. Use Power BI as a recurring monitoring system: management can review the same KPI framework regularly and drill into the segment causing a change.

10. Conclusion

The ElectroHub project demonstrates how a Data Analyst can move from raw transactional data to actionable business decisions. The value of the project is not the dashboard alone; it is the complete analytical process behind it.

Data cleaning created a dependable foundation by addressing duplicates, missing or inconsistent information, data types, suspicious values, and business-rule issues. Python EDA then established the key performance indicators and patterns that deserved deeper investigation. SQL transformed those questions into structured, repeatable analyses across the relational data model. Finally, Power BI converted the results into an interactive management reporting system.

The project also demonstrates an important business principle: increasing sales is not enough. Sustainable e-commerce growth requires a balance between revenue, profit margin, customer value, marketing efficiency, and operational reliability. By connecting these dimensions, ElectroHub's management can identify where growth is profitable, where customer value is strongest, which products require margin attention, and where delivery or service problems need operational action.

Therefore, the final outcome is a decision-support framework rather than a collection of charts: measure performance → identify the driver → investigate the problem → take action → monitor the result.

11. Interview-Ready Project Summary

I developed an end-to-end E-commerce Sales and Business Analytics project for ElectroHub to understand how the company could grow sales and profit while improving customer retention and operational performance.

The dataset contained five related tables: customers, orders, order_items, products, and delivery. I first performed data profiling and cleaning using Excel/Power Query and Python. I checked missing values, duplicates, data types, categorical inconsistencies, invalid values, outliers, and relationships between the tables. For outliers, I used business logic instead of automatically deleting extreme values—for example, a genuine late delivery should remain visible because it is an operational issue.

After cleaning, I used Python for EDA and focused on business KPIs such as total sales, total profit, profit margin, orders, AOV, category performance, customer segments, acquisition channels, and delivery performance. I then used SQL to answer structured questions such as monthly sales, category profitability, top products, customer-tier performance, acquisition-channel performance, repeat customers, and late-delivery rate.

For the final reporting layer, I built a four-page Power BI dashboard: Executive Overview, Customer & Growth Analysis, Product & Profitability Analysis, and Operations & Delivery. I created reusable DAX measures for Sales, Profit, Orders, Customers, AOV, Profit Margin, Repeat Customers, Late Delivery %, Cancellation Rate, Return Rate, and delivery/shipping metrics.

The main business value of the project is that it connects revenue with profitability and customer and operational drivers. Instead of only showing that a category or product sells well, the analysis asks whether it is profitable; instead of only counting customers, it examines repeat behavior and acquisition-channel value; and instead of only reporting delivery time, it compares actual performance with promised delivery.

Based on the analysis, the key recommendations are to prioritize profitable products, optimize discounts, invest in high-value acquisition channels, improve customer retention, and investigate warehouses or states with weaker delivery performance.

12. Skills Demonstrated

Area	Skills Demonstrated
Data Cleaning	Missing-value checks, duplicates, data types, text standardization, validation, outlier investigation, business-rule checks
Python	Pandas aggregation, grouping, distinct counts, KPI calculations, EDA, business-focused analysis
SQL	JOINS, GROUP BY, aggregate functions, DISTINCT, CASE, HAVING, subqueries, date-based analysis
Power BI	Data modeling, interactive visuals, slicers, cards, dashboard layout, drill-down style analysis
DAX	SUM, DISTINCTCOUNT, DIVIDE, CALCULATE, FILTER, VALUES, conditional measures
Business Analysis	KPI selection, profitability analysis, customer segmentation, acquisition analysis, operational analysis
Communication	Converting technical analysis into business problems, insights, and recommendations